

Labeling Guidelines: Mining Site Video Object Tracking

Project: Multi-class object tracking on aerial mining/quarry operations footage

Annotator: Tshepang Lumatha

Tool: Label Studio (Video Object Tracking template)

Source footage: 18-second aerial drone clip of an active mining/quarry site

1. Purpose

This document defines the labeling rules used to annotate vehicles, equipment, and personnel in aerial mining site footage. The goal is to produce consistent, model-ready tracking data suitable for use cases such as equipment utilization monitoring, site safety analysis, and fleet tracking.

2. Class Definitions

Class	Definition
Wheel Loader	Rubber-tired loader equipment with a front bucket, used for moving stockpiled material.
Dump Truck	Rigid or articulated haul truck used for transporting bulk material on-site.

Tanker Truck	Truck with a cylindrical tank body (water, fuel, or liquid transport).
Pickup Truck	Light passenger/utility vehicle (personnel transport, site supervision).
Flatbed Trailer	Trailer or lowboy used to transport equipment (e.g., excavators) rather than bulk material.
Excavator	Tracked or wheeled excavation equipment, whether actively working or being transported.
Worker	Any visible person on foot within the site boundary.

Not labeled: Stationary site infrastructure (conveyors, processing structures, stockpiles, dust/smoke) — these are environmental context, not trackable objects.

3. Bounding Box Rules

- Each individual vehicle/equipment/person gets its own bounding box — never group multiple objects into a single box, even if closely parked or moving together.
- Boxes should tightly hug the visible extent of the object, including any load being carried (e.g., a flatbed trailer's box includes the equipment it's transporting only if the equipment isn't independently trackable/separately classed).
- Given the aerial/drone perspective, boxes are drawn from a top-down view — "tight" means matching the visible silhouette from above, not an assumed ground-level profile.

4. Occlusion Rule

An object is tracked (bounding box maintained) if at least 40% of it remains visible.

- If an object is briefly fully obscured by dust/smoke plumes (a known feature of this footage) for **under 1 second**, the same track ID is preserved across the gap — treat it as a continuous track, not a new object.
- If an object is obscured for **longer than 1 second** or exits the dust cloud in a materially different position suggesting ambiguity, a note is added flagging the uncertain span for review, and the ID is only continued if there's high confidence it's the same object (matching size, direction of travel, and position trajectory).

5. Re-Identification Rule

- If a vehicle exits the frame and later re-enters, it is assigned a **new track ID**, unless its exit and re-entry are close enough in time and position to make it unambiguous that it's the same vehicle (e.g., briefly clipped by frame edge).
- This default is chosen because aerial footage at this scale makes confident re-identification difficult without additional distinguishing features (license plates, unique markings), and a conservative rule reduces the risk of incorrectly merging two different vehicles into one track.

6. Class Ambiguity Rules

- **Excavator on a flatbed trailer:** labeled as a separate "Excavator" bounding box nested within/near the "Flatbed Trailer" box, since it is a distinct trackable piece of equipment, not part of the trailer itself.
- **Parked vs. active equipment:** no distinction is made in the class label between parked and actively operating equipment (e.g., a stationary wheel loader is still "Wheel Loader"). Operational state is considered out of scope for this tracking task.

- **Unclear vehicle type** (e.g., partially obscured or too small to confidently classify): labeled with the best-fit class based on visible shape/size, with a note flagging low confidence.
-

7. Known Challenges in This Footage

- **Dust/smoke plume** near the center of the site periodically reduces visibility of nearby vehicles and the processing equipment.
 - **Aerial perspective** makes vehicle classes with similar top-down silhouettes (e.g., dump truck vs. flatbed trailer) harder to distinguish at a glance — resolved by looking at load shape and cab position.
 - **Short clip length (18 seconds)** limits the number of long-duration tracks but keeps the sample focused and fully reviewable.
-

8. Export & Documentation Notes

- Annotations exported in Label Studio's native JSON format.
- This guidelines document is included alongside the exported annotations and sample screenshots as part of the portfolio submission, to demonstrate labeling consistency and documented decision-making not just raw output.